

Unit V: Electromagnetic waves

Chapter–8: Electromagnetic Waves

GIST OF CHAPTER:

Basic idea of displacement current, Electromagnetic waves, their characteristics, their transverse nature (qualitative idea only).

Electromagnetic spectrum (radio waves, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays) including elementary facts about their uses.

CONTENT/ CONCEPTS-

Basic idea of displacement current

Conduction Current (i_c)- Conduction current is the current, which arises due to flow of electrons through the connecting wires in an electric circuit.

Displacement current (i_d)- Displacement current is the current, which arises due to time rate of change of electric flux (ϕ_E).

Flaw in Ampere's circuital law

When a capacitor is charged or a charged capacitor is allowed to discharge,

To find the magnetic field at a point such as P, in a region outside the parallel plate capacitor using **Ampere's circuital law**.

- (I) If a plane circular loop of radius r whose plane is perpendicular to the direction of the current-carrying wire, and which is centred symmetrically with respect to the wire. Then magnetic field at point P is given by

$$B(2\pi r) = \mu_0 i_{\text{passing through the surface}} \Rightarrow B = \frac{\mu_0 i}{2\pi r}$$

- (II) If a different surface, which has the same boundary which is a pot like surface has its bottom between the capacitor plates which nowhere touches the current. Hence current passing through surface = 0.

Then magnetic field at point P is given by

$$B(2\pi r) = \mu_0 i_{\text{passing through the surface}} \Rightarrow B = \frac{\mu_0 \times 0}{2\pi r} = 0$$

So we have a *contradiction*; calculated one way, there is a magnetic field at a point P; calculated another way, the magnetic field at P is zero.

Maxwell's correction in Ampere's circuital law

The second surface has a surface S between the capacitor plates. If the plates of the capacitor have an area A, and a total charge Q, the magnitude of the electric field E between the plates is

$$E = \frac{Q}{\epsilon_0 A}$$

The field is perpendicular to the surface S. It has the same magnitude over the area A of the capacitor plates, and vanishes outside it.

The electric flux through the surface S Using Gauss's law,

$$\phi_E = EA = \frac{Q}{\epsilon_0} \Rightarrow Q = \epsilon_0 \phi_E \Rightarrow \frac{dQ}{dt} = \epsilon_0 \frac{d\phi_E}{dt}$$

Here $\frac{dQ}{dt} = \epsilon_0 \frac{d\phi_E}{dt} = i_d$ is called **Displacement current**.

The displacement current between plates of capacitor is always equal to the conduction current in wire.

i_c = Conduction current(flows in conducting wire)

i_d = Displacement current (between plates of capacitor).

Hence, the Ampere's circuital law is modified.

Modified Ampere's circuital law states that the line integral of magnetic field B over a closed boundary of an open surface is equal to μ_0 , times the sum of the conduction current (I_c) and the displacement current (I_d) threading the surface. Mathematically-

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_c + \epsilon_0 \frac{d\phi_E}{dt})$$

It is also known as **Ampere-Maxwell's circuital law**.

Maxwell's Equations:- Following four equations, which describe the laws of electromagnetism, are called Maxwell's equations

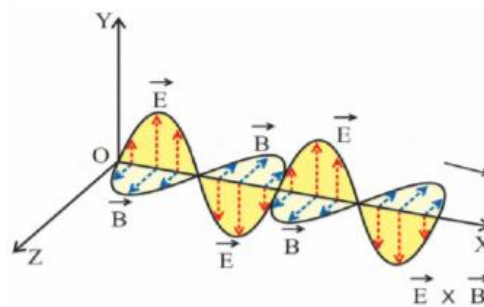
- (i) $\oint \mathbf{E} \cdot d\mathbf{S} = q/\epsilon_0$ (Gauss's law in electrostatics)
- (ii) $\oint \mathbf{B} \cdot d\mathbf{S} = 0$ (Gauss's law in magnetism)
- (iii) $\oint \mathbf{E} \cdot d\mathbf{l} = -d\phi_B/dt$ (Faraday's law of electromagnetic induction)
- (iv) $\oint \mathbf{E} \cdot d\mathbf{l} = \mu_0(I_c + I_d)$ (Ampere-Maxwell's circuital law)

Source of electromagnetic waves-

Electromagnetic waves are produced due to oscillating or accelerating charge.

Characteristics of Electromagnetic Waves

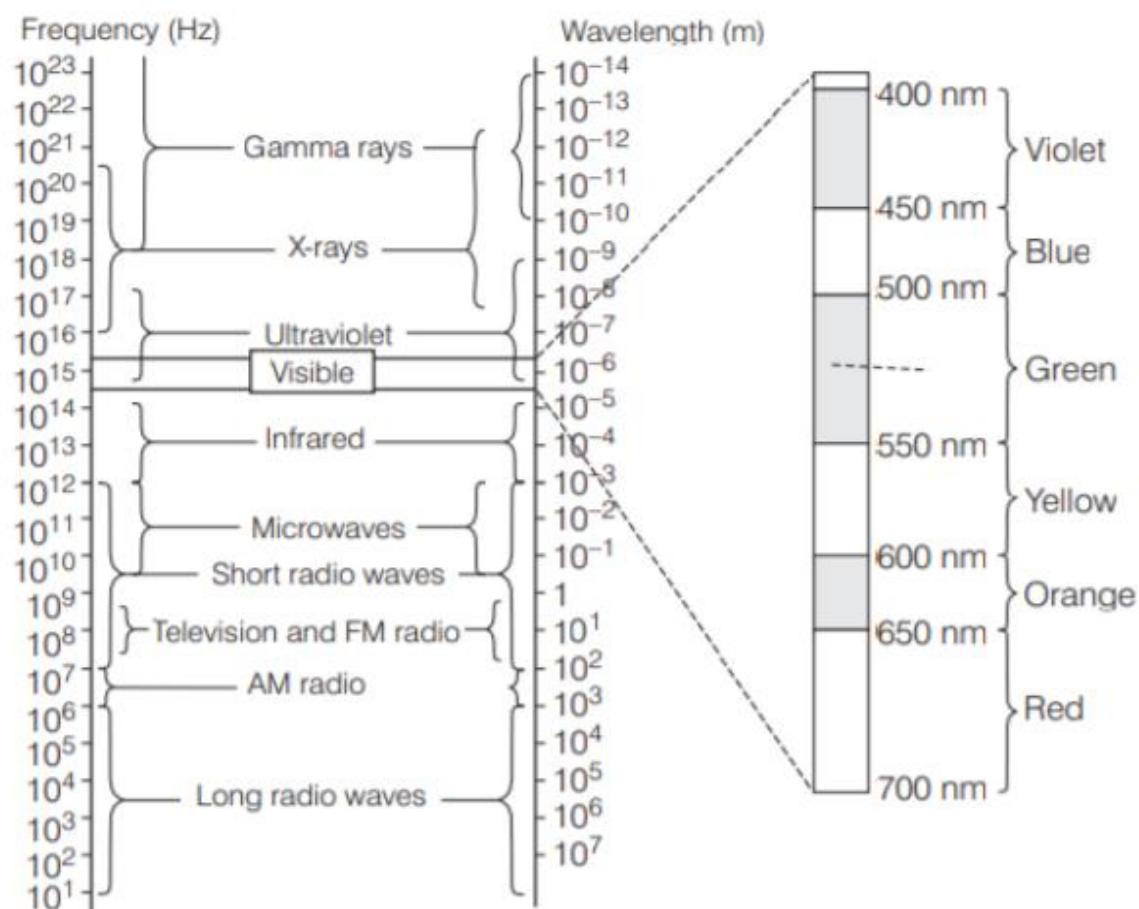
1. The **electric (E)** and **magnetic fields (B)** vary sinusoidally in space and time and propagate through space, such that the two fields are perpendicular to each other and perpendicular to the direction of propagation, constitute electromagnetic waves.



2. The direction of propagation of an electromagnetic wave is given by the cross product of electric field and magnetic field vectors
3. The electromagnetic waves are transverse in nature.
4. The velocity of electromagnetic waves in free space is given by $c = 1/\sqrt{\mu_0\epsilon_0} = 3 \times 10^8 \text{ m/s}$. In a material medium, velocity of electromagnetic waves is given by $v = 1/\sqrt{\mu\epsilon}$
5. The ratio of the amplitudes of electric and magnetic fields is constant and it is equal to velocity of the electromagnetic waves in free space. Mathematically- $E_0/B_0 = c$
6. The energy in electromagnetic waves is divided equally between the electric and magnetic field vectors
7. The electric vector of an electromagnetic wave is responsible for its optical effect. For this reason, the electric vector is also called light vector. The energy transported by electromagnetic waves is given by $U = hv = hc/\lambda$ where U is energy transported by electromagnetic waves in a given time and c is speed of electromagnetic waves in free space
8. The momentum transported by electromagnetic waves is given by $p = U/c = hv/c = h/\lambda$.
9. The intensity of electromagnetic waves i.e. energy crossing per second per unit area of a surface normally is given by $I = \frac{1}{2}\epsilon_0 E_0^2$
10. When electromagnetic waves strike a surface, they exert pressure on the surface.
11. The orderly distribution of electromagnetic waves (according to wavelength or frequency) in the form of distinct groups, having widely differing properties, is called electromagnetic spectrum.
12. The main parts of electromagnetic spectrum are namely- **γ -rays, X-rays, ultra-violet rays, visible light, infrared rays, microwaves and radio waves.**
13. The frequency of electromagnetic waves is its inherent characteristic. When an electromagnetic wave travels from one medium to another, its wavelength changes but frequency remains unchanged.
14. All the types of electromagnetic waves travel with the same speed in free space.
15. The orderly arrangement of EM waves in increasing or decreasing order of wavelength and frequency is called electromagnetic spectrum.

Electromagnetic wave spectrum

Electromagnetic wave spectrum is shown below



[Hint to remember EMW – Rahul's Mother Is Visiting Uncle Xavier Garden.]

Elementary facts about the Uses of Electromagnetic waves

Radio waves - Radio waves are produced by the accelerated motion of charges in conducting wires. They are used in radio and television communication systems. They are generally in the frequency range from 500kHz to about 1000MHz. The AM (amplitude modulated) band is from 530kHz to 1710kHz. Higher frequencies upto 54MHz are used for short wave bands. TV waves range from 54MHz to 890MHz. The FM (frequency modulated) radio band extends from 88MHz to 108MHz. Cellular phones use radio waves to transmit voice communication in the ultrahigh frequency (UHF) band.

Microwaves-Microwaves (short-wavelength radio waves), with frequencies in the gigahertz (GHz) range, are produced by special vacuum tubes (called klystrons, magnetrons and Gunn diodes). Due to their short wavelengths, they are suitable for the radar systems used in aircraft navigation. Radar also provides the basis for the speed guns used to time fast balls, tennis serves, and automobiles. Microwave ovens are an interesting domestic application of these waves. In such ovens, the frequency of the microwaves is selected to match the resonant frequency of water molecules so that energy from the waves is transferred efficiently to the kinetic energy of the molecules. This raises the temperature of any food containing water.

Infrared waves-Infrared waves are produced by hot bodies and molecules. This band lies adjacent to the low-frequency or long-wave length end of the visible spectrum. Infrared waves are sometimes referred to as heat waves. This is because water molecules present in most materials readily absorb infrared waves (many other molecules, for example, CO_2 , NH_3 , also absorb infrared waves). After absorption, their thermal motion increases, that is, they heat up and heat their surroundings. Infrared lamps are used in physical therapy. Infrared radiation also plays an important role in maintaining the earth's warmth or average temperature through the greenhouse effect. Incoming visible light (which passes relatively easily through the atmosphere) is absorbed by the earth's surface and reradiated as infrared (longer wavelength) radiations. This radiation is trapped by greenhouse gases such as carbon dioxide and water vapour. Infrared detectors are used in Earth satellites, both for military purposes and to observe growth of crops. Electronic devices (for example semiconductor light emitting diodes) also emit infrared and are widely used in the remote switches of household electronic systems such as TV sets, video recorders and hi-fi systems

Visible rays-It is the most familiar form of electromagnetic waves. It is the part of the spectrum that is detected by the human eye. It runs from about 4×10^{14} Hz to about 7×10^{14} Hz or a wavelength range of about 700–400 nm. Visible light emitted or reflected from objects around us provides us information about the world. Our eyes are sensitive to this range of wavelengths. Different animals are sensitive to different range of wavelengths. For example, snakes can detect infrared waves, and the 'visible' range of many insects extends well into the ultraviolet.

Ultraviolet rays-It covers wavelengths ranging from about 4×10^{-7} m (400 nm) down to 6×10^{-10} m (0.6 nm). Ultraviolet (UV) radiation is produced by special lamps and very hot bodies.

The sun is an important source of ultraviolet light. But fortunately, most of it is absorbed in the ozone layer in the atmosphere at an altitude of about 40 – 50 km. UV light in large quantities has harmful effects on humans. Exposure to UV radiation induces the production of more melanin, causing tanning of the skin. UV radiation is absorbed by ordinary glass. Hence, one cannot get tanned or sunburn through glass windows. Welders wear special glass goggles or face masks with glass windows to protect their eyes from large amount of UV produced by welding arcs. Due to its shorter wavelengths, UV radiations can be focussed into very narrow beams for high precision applications such as LASIK (Laser assisted in situ keratomileusis) eye surgery. UV lamps are used to kill germs in water purifiers. Ozone layer in the atmosphere plays a protective role, and hence its depletion by chlorofluorocarbons (CFCs) gas (such as freon) is a matter of international concern.

X-rays -Beyond the UV region of the electromagnetic spectrum lies the X-ray region. We are familiar with X-rays because of its medical applications. It covers wavelengths from about 10^{-8} m (10 nm) down to 10^{-13} m (10^{-4} nm). One common way to generate X-rays is to bombard a metal target by high energy electrons. X-rays are used as a diagnostic tool in medicine and as a treatment for certain forms of cancer. Because X-rays damage or destroy living tissues and organisms, care must be taken to avoid unnecessary or over exposure.

Gamma rays -They lie in the upper frequency range of the electromagnetic spectrum and have wavelengths of range from about 10^{-10} m to less than 10^{-14} m. This high frequency radiation is produced in nuclear reactions and also emitted by radioactive nuclei. They are used in medicine to destroy cancer cells. As mentioned earlier, the demarcation between different region is not sharp and there are over laps.

Type	Wavelength range	Production	Applications
Radio	>0.1 m	Rapid acceleration and decelerations of electrons in aerials	In radio and T.V. communication system, Cellular phones
Microwave	0.1m to 1mm	Klystron, magnetron valve, Gun diode	In aircraft and satellite communication and to cook food (microwave oven)
Infrared	1mm to 700mm	Vibration of atoms and molecules	In Physiotherapy, green house, military purpose and in agriculture

Visible	700 nm to 400nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	In fibre optic communications, photography.
Ultraviolet	400nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	In welding , eye surgery and to kill germs
X-rays	1nm to 1pm	X-ray tubes or inner shell electrons of atom	To destroy living tissues and organisms, checking for bone fractures.
Gamma rays	<1pm	Radioactive decay of the nucleus	To destroy cancer cells

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